

Use Case Specification: Book Event Ticket

1. Overview

- **Use Case ID:** UC-01
- **Use Case Name:** Book Event Ticket
- **Primary Actor:** Attendee

2. Preconditions and Postconditions

Preconditions:

- The attendee is logged into their account.
- The event is published, active, and has available tickets.

Postconditions:

- The ticket inventory is decremented by the purchased quantity.
- A digital ticket with a unique ID is issued to the attendee.
- A confirmation email with the ticket receipt is sent.

3. Main Success Flow

1. Attendee searches or browses active events and selects an event.
2. Attendee selects ticket quantity and proceeds to checkout.
3. System verifies seat availability and reserves the requested tickets.
4. Attendee enters payment details and confirms the order.
5. System processes payment through the external payment gateway.
6. System deducts the purchased tickets from remaining event capacity.
7. System generates a digital ticket with a unique ID and QR code.
8. System displays a confirmation screen and dispatches an email receipt with the ticket attached.

4. Alternate Flow: Payment Authorization Fails

Branches from Step 5 of the Main Flow when payment is declined.

- 5a.** The payment gateway rejects the transaction.
- 5b.** System releases the reserved tickets back to the available inventory.
- 5c.** System displays an error message explaining that payment was unsuccessful.
- 5d.** Attendee is prompted to retry with different payment details or cancel checkout.